

NULL Values & 3-Valued Logic

Q. Explain the concept of NULL in SQL. How do we check for NULL values, handle them in expressions, and how do they affect comparisons and aggregate functions?

> **Definition to Remember**

> **NU**
an em
NULL

1. Key Rules About NULL

- **NULL 0** (zero is a number; NULL is the absence of a value)

- **N**

2. Checking for NULL Values

You **cannot** use `= NULL` or `!= NULL` these always return UNKNOWN. Use specialized operators:

```
```sql
```

-- IS N

-- IS NOT NULL: find rows where a column has a value

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### 3. Three-Valued Logic

SQL uses True, False, and **Unknown** (instead of just True/False):

Condition	Result

| :--- | :

### 4. Handling NULL with COALESCE()

`COALESCE(val1, val2, ...)` returns the **first non-NULL value** in the list used to substitute a default value for NULLs.

```
```sql
```

5. Effect on Aggregate Functions

- `SUM()`, `AVG()`, `MAX()`, `MIN()` **automatically ignore NULL rows**.

- `CO`

Example: Salaries = [100, 200, NULL] `AVG(Salary)` = (100+200)/2 = **150** (NULL ignored)

> **Must-Write Points (for 10 marks)**

> 1. N

> **Quick Recall**

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